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📍 NYCU, Hsinchu, Taiwan

EDUCATION

Master of Science, Structural and Material Engineering

National Yang Ming Chiao Tung University, Taiwan | Aug 2025 – Expected Jun 2027

Average score (1st Sem.): - 4.27/4.3 ([Received Outstanding Student Award](#))

Bachelor of Engineering, Civil Engineering

Aligarh Muslim University, India | Aug 2021 – Jun 2025

Percentage: 86.61% ([Under top 10 rank in a class of 48 students](#))

RESEARCH INTERESTS

Composites | Computational Geomechanics | Machine Learning | Soil-Structure Interaction

RESEARCH EXPERIENCE

- **Physics Informed Neural Network for Elastodynamics Green's Function** (Sep 2025–Present)
Research Project | Advisor: Chair Professor Ernian Pan, National Yang Ming Chiao Tung University.
 - Developing and implementing a Physics-Informed Neural Network (PINN) framework to evaluate elastodynamic Green's functions, providing efficient computational solutions for dynamic Soil-Structure Interaction problem.
- **Advanced Neural Operator Modeling for Multilayered Isotropic Media** (Sep 2025–Present)
Research Project | Advisor: Chair Professor Ernian Pan, National Yang Ming Chiao Tung University.
 - Applying advanced deep learning architectures, specifically complex-valued Residual Networks and Fourier Neural Operators (FNOs), to resolve wave propagation and elastodynamic behaviors in isotropic half-space multilayered media.
- **Physics-Guided Machine Learning for Pile Uplift Capacity Prediction** (Jan 2026– July 2026)
Research Project | Advisor: Chair Professor Ernian Pan, National Yang Ming Chiao Tung University. ([Link](#))
 - Designing a physics-guided machine learning framework utilizing multilayered, model-agnostic techniques to interpret underlying mechanical behaviors of soil pile interactions and accurately predict the uplift capacity of piles foundation.
- **Explainable AI for strength and mechanistic Interpretation of stabilized soil** (Jan 2026–June 2026)
Research Project | Advisor: Chair Professor Ernian Pan, National Yang Ming Chiao Tung University. ([Link](#))
 - Developed multiple machine learning models and multilayered explainability techniques to explore the mechanistic properties of lime- and cement-stabilized soil and accurately predict its strength.
- **Soil Stabilization and Rockfall Hazard Prediction** (July 2025 – August 2025)
Internship | Advisor: Dr. Riya Bhowmik, Indian Institute of Technology, J&K, India ([Link](#))
 - Conducted comprehensive experimental soil testing to evaluate the efficacy of various additives in enhancing soil stability and mechanical properties.
 - Predicted rockfall parameters by integrating RocFall 2.0 simulations with AI models.
- **Utilizing the borosilicate glass (BG), PVA and SCMs in Concrete** (Aug 2024 – June 2025)
Bachelor Thesis | Advisor: Prof. Rehan & Prof. Hasan, Aligarh Muslim University. ([Link](#)) ([Certificate](#))
 - We are using BG powder (10-20%), silica fume (5-15%) and PVA fiber in different percentage in concrete.
 - Part of the work was presented at the ASCE conference, and the remaining experiments are ongoing.
- **Advancing Green Construction through Alkali Treated Abaca Fibers and Zeolite Integration** (Jul 2024 – Feb 2025)
Research Project | Advisor: Prof. Rizwan Ahmad, Aligarh Muslim University. ([Link](#))
 - Zeolite was used as a partial replacement for cement and Abaca fiber (0.25-1.5%) was used as reinforcement.
 - Found maximum strength at 7.5% zeolite and 0.5% abaca fiber.
- **Insights into Road Safety: Literature Review and Comprehensive Data Analysis Approach** (Dec 2024 – Jan 2025)
Internship | Advisor: Chief Scientist: A Mohan Rao, Council of Scientific and Industrial Research-CRRI, India ([Link](#))
 - Analyzed accident data to identify patterns and develop ML models for accident severity prediction.
- **Black-Box to Explainable AI: Predicting Compressive Strength of UHPC** (May 2024 – Oct 2024)
Research Project | Advisor: Dr. Syed Muhammad Ibrahim, Aligarh Muslim University. ([Link](#))
 - Developed ML models (DT, RF, GB, AB) to predict Ultra High Performance Concrete (UHPC) compressive strength.

- Applied SHAP plots for feature analysis and model interpretability.
- Provided insights into ML as a viable alternative to traditional UHPC testing methods.
- **Leveraging AI to Unlock the Potential of Sustainable Fiber-Reinforced Concrete Materials** (May 2024 – Nov 2024)
Research Project | Advisor: Dr. Syed Muhammad Ibrahim, Aligarh Muslim University. ([Link](#))
 - Designed GB, RF, and ANN models with 1,500 experimental instances.
 - Conducted feature impact analysis to enhance model reliability.
 - Achieved optimal performance with GB and RF, validated through feature impact analysis.
- **Comparative Analysis of Machine Learning Models for Predicting Split Tensile Strength** (Dec 2023–Apr 2024)
Research Project | Advisor: Dr. Syed Muhammad Ibrahim, Aligarh Muslim University. ([Link](#))
 - Developed and compared ensemble ML algorithms for SCM-blended lightweight concrete.
 - Achieved high predictive accuracy using Gradient Boosting, validated by MSE, RMSE, and R².

PUBLICATIONS ([Research Gate](#))

- Explainable AI for Strength Prediction and Mechanistic Interpretation of Lime and Cement Stabilized Soils- Journal of Geotechnical & Geological Engineering- *A Azeem, SS Ansari, E Pan, CP lin.* ([10.1007/s10706-026-03720-7](https://doi.org/10.1007/s10706-026-03720-7))
- Advancing Green Construction through Alkali Treated Abaca Fibers and Zeolite Integration – Journal of building Engineering -*A Azeem, JN Akhtar, RA Khan.* (<https://doi.org/10.1016/j.jobbe.2026.115215>)
- Comparative Analysis of Conventional and Ensemble Machine Learning Models for Predicting Split Tensile Strength in Stressed SCM-Blended Lightweight Concrete- *SS Ansari, A Azeem, M Asad, k Zafar, SM Ibrahim* (<https://doi.org/10.1016/j.matpr.2024.04.081>)
- Black-Box to Explainable AI: Predicting Compressive Strength of Ultra-High Performance Concrete – ASCE International Conference (Accepted, Paper Id 118)
- Leveraging AI to Unlock the Potential of Sustainable Fiber-Reinforced Concrete Materials – ASCE International Conference (Accepted, Paper Id 189)
- Performance Evaluation of Concrete Containing Borosilicate Glass Powder and Silica Fume – ASCE International Conference (Accepted, Paper Id 303)

PROFESSIONAL EXPERIENCE

- Seed Teaching Assistant, TSIR-National Institute of Applied Research, Taiwan** (July 2026–Present)
 - Assisting with educational camps, workshops, and science outreach programs, as well as facilitating global exchanges.
- Graduate Research Assistant, National Yang Ming Chiao Tung University, India** (Sept 2025–Present)
 - Developing advanced machine learning models and semi-analytical approaches to analyze soil-structure interactions.
- Undergraduate Research Assistant, Aligarh Muslim University, India** (Aug 2024–June 2025)
 - Executed comprehensive experimental testing to support the development and structural evaluation of sustainable concrete.
- Site Engineer, Dwarika Infocom Pvt. Ltd., Noida, India** (Jan 2021–Jun 2021)
 - Supervised field operations ensuring adherence to design, quality, and safety standards.
- Intern, John Holland, Australia** (Feb 2024–Mar 2024)
 - Calculated concrete volumes for bridge pile caps and abutments; compared subcontractor rates.
 - Evaluated building drawings to streamline task sequencing and improve project efficiency.

SKILLS

- **Software:** STAAD Pro, AutoCAD, Revit, Matlab, Plaxis, Ansys
- **Programming language:** Python, R, Html
- **Material Testing:** UPV test, Electrical resistivity, SEM analysis, Triaxial shear, Proctor compaction, Direct shear test

AWARDS AND ACHIEVEMENTS

- **2026:** Excellent international volunteer awards for serving on NYCU International day.
- **2026:** Honor student distinction award for outstanding performance in 1st semester of MS.
- **2026-2027:** Outstanding existing student scholarship award for maintaining outstanding grades by NYCU.
- **2024:** Merit-cum-Means scholarship by Bharat Sewa Trust.
- **2021-2025:** Merit-cum-Means scholarship for Bachelor of Engineering by State government.
- **2017 – 2020:** Merit-cum-Means scholarships (3 consecutive years) by Central government.
- **2014:** 1st Prize, Quiz Competition sponsored by 3S classes.
- **2013:** 2nd Prize, Essay Writing Competition.

WORKSHOPS & TRAINING

- Presented three research paper at the ASCE International Conference *PEC, India (Feb 2025)*
- CFD simulation in Civil and Environmental Engineering *ACI, America (Mar. 7,2024)*
- Workshop on Disaster Risk Reduction in Civil Engineering Structures *AMU, India (Oct 2022)*
- Short-term Training on Disaster Management & Mitigation *NIDM & ZHCET, AMU (Nov 2022)*

VOLUNTEER WORK

- Served as a event planning team member of International Student Association, NYCU for fall semester.
- Served as a National Cadet Corps (NCC) member for three consecutive years.
- Contributed as a Coordinator and volunteer in the NGO, Wing of Desire for 1 year.
- Contributed as a volunteer in the NGO, Aligarh Renaissance for 1 year.

CO-CURRICULAR ACTIVITIES

- Led a group of 10 students as the group leader in a survey camp, in Amu, India
- Participated in the 'Eco Fox Run' to promote a clean and green environment, conducted by the Eco Club Amu, India.
- Qualified for the National Muslim Talent Search Examination.

MOOCs

- Artificial Intelligence applications for Civil Engineers. *(Udemy, Nov. 8,2024)*
- 100 days of Code: The complete Python Pro bootcamp. *(Udemy, Apr. 8, 2024)*